

MUFFIN for the jet $\rightarrow \tau_h$ background

v29pre, 2017 – status and validation

What MUFFIN is

MULTivariate Fake-Factor INFERENCE (Andreou, Colling, Winterbottom; CMS-PAS-TAU-25-001)

The binned fake factor is replaced by a continuous, multi-dimensional per-object weight:

$$w_{\text{MUFFIN}}(z) = \frac{p_{\text{pass}}^{\text{data}}(z) - p_{\text{pass}}^{\text{sim}}(z)}{p_{\text{fail}}^{\text{data}}(z) - p_{\text{fail}}^{\text{sim}}(z)}$$

- ▶ This is *the same quantity* our map delivers as $FR/(1 - FR)$: the denominator is inclusive-loose, so $FR/(1 - FR) = N_{\text{pass}}/N_{\text{fail}}$. Only the estimation changes – continuous in 7 variables instead of $p_T \times |\eta|$ cells per prong.
- ▶ Estimated with a single BDT classifier trained on pass vs. fail with **signed** weights (+1 data, $-w_{\text{ana}}$ genuine- τ simulation). The stationary point of that weighted log-loss is $s = A/(A+B)$ over the *net* densities, so $w_{\text{MUFFIN}} = s/(1 - s) = e^{\text{margin}}$ exactly.
- ▶ Inputs are the poster's list verbatim: τ_h decay mode, $p_T^{\text{jet}}/p_T^\tau$, p_T^τ , N_{jets} , N_b , η , ϕ . No mother-jet flavour axes (dropped from our FR method).

Five components, following the poster; 30 trainings in total (1 nominal + 20 bootstrap + 7 hyper-parameter + 2 background-subtraction):

component	how	value ($1\sigma_{\text{VR}}$)
statistical	20 Poisson-bootstrap retrainings	2.4%
modelling	7 hyper-parameter variations	1.2%
bkg. subtraction	$\pm 10\%$ on the subtracted genuine- τ MC	1.8%
non-closure + extrapolation	VR closure residual	2.5%
total	quadrature	4.0%

Per bin of p_T^τ , H_T and p_T^{jet1} the MUFFIN band is a flat **11.4%**, while the binned map grows from 13.7% to **16.5%** into the tails, where its cells run out of events.

Validation I: does the fake factor travel between channels?

$t\bar{t}$ MC truth, fake τ_h only – no data-driven subtraction needed

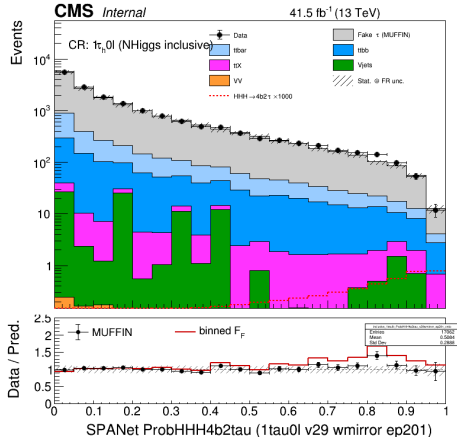
Measure the fake factor on $t\bar{t}$ MC in $1\tau_h1l$, predict $1\tau_h0l$. 724k / 270k fake τ_h , so the transfer is tested well below a percent.

	integral	p_T^τ	p_T^{jet}	$ \eta $	N_{jets}	p_T^{miss}
<i>transfer</i> $1\tau_h1l \rightarrow 1\tau_h0l$						
MUFFIN	1.018	1.8%	3.3%	2.6%	1.9%	3.5%
binned map	1.031	14.8%	3.0%	3.4%	4.1%	5.0%
inclusive-only	1.046	8.8%	16.9%	4.9%	4.6%	6.4%
<i>reference</i> $1\tau_h0l \rightarrow 1\tau_h0l$, out-of-fold						
MUFFIN	1.000	0.5%	4.1%	5.2%	0.6%	2.4%
binned map	1.000	16.2%	0.5%	1.7%	3.2%	2.6%

The $t\bar{t}$ fake factor transfers well: the inclusive value differs by 4.5% between channels, and a differential fake factor absorbs most of it – **1.8% residual for MUFFIN**, 3.1% for the binned map, 4.6% for a constant. MUFFIN transfers *better*, because the channels differ largely in N_{jets}/p_T and it is differential in those.

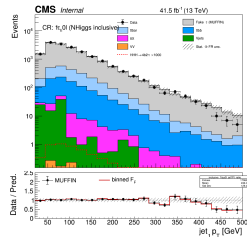
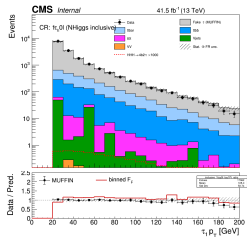
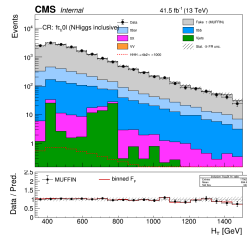
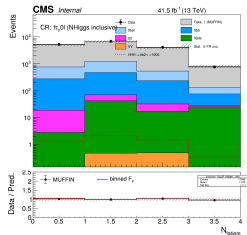
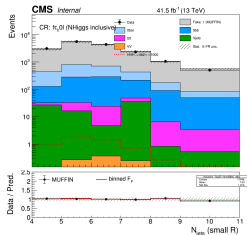
Validation II: overall closure is the same, the score region is better

$1\tau_h 0l$ inclusive – black points MUFFIN, red line the binned F_F



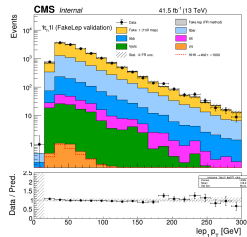
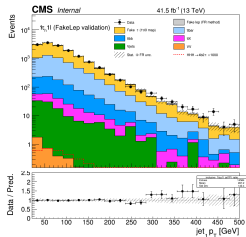
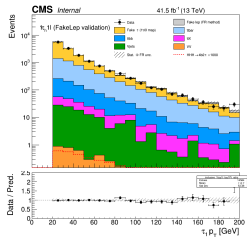
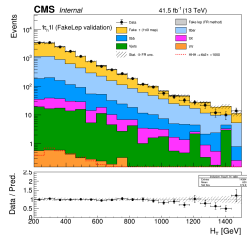
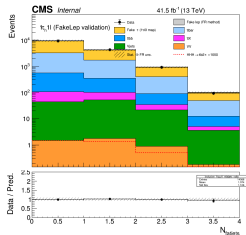
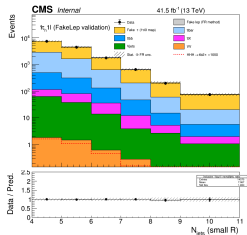
- ▶ Same events, same MC-prompt stack, same anti-ID selection and subtraction – only the fake factor differs, so the gap between the two curves *is* the difference between the methods.
- ▶ Integrated yields agree to **0.02%** ($1\tau_h 0l$ VR) and **0.08%** (inclusive): MUFFIN does not change the normalisation.
- ▶ It changes the **shape**. In the high-score region the binned F_F drifts up to ~ 1.4 , while MUFFIN stays within the band.
- ▶ Over every SPANet ProbHHH4b2tau training the closure improves by **1.4–2.7%**; p_T^τ improves from 11.9% to **3.8%**.

Fake τ from MUFFIN; ratio pad: points MUFFIN, red line binned F_F



$1\tau_h 1l$ inclusive

Fake τ from MUFFIN measured *in-channel*; fake leptons from the data-driven lepton FR



The $1\tau_h 1l$ fake factor is measured in-channel

Measured in the channel's own MR (`mr60: jet4DeepFlavB < 0.035`, the balanced 60/40 split of the fake FO pool), so the agreement on the previous slide is *not* closing by construction – an inclusive in-channel fit would reproduce the tight region trivially.

- ▶ pool: `kind_category_FR==1` with a tight lepton, SingleMuon + SingleElectron, 27060 data events in the MR
- ▶ inclusive F_F : 0.3357 (MR) vs 0.3393 (VR), 1.1% apart
- ▶ feature ranking as in $1\tau_h 0l$: $p_T^{\text{jet}}/p_T^\tau \gg \text{decay mode} > \eta$

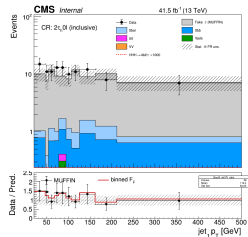
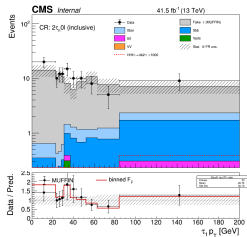
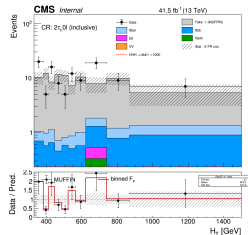
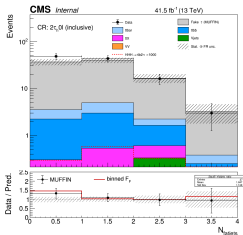
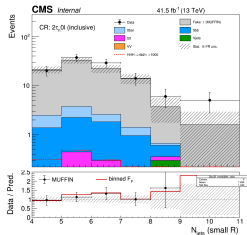
class	out-of-fold MR	MR $\rightarrow$ VR
nj45 1p	0.989	0.993
nj45 3p	1.032	0.977
nj6 1p	1.041	0.937
nj6 3p	1.083	0.972
total	1.008	0.975

Left:

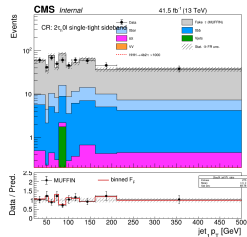
every event predicted by a model that never saw it.
Right: measured in the MR, applied to the VR – the actual validation.

$2\tau_h 0l$ inclusive

Fake τ from MUFIN, measured in $1\tau_h 0l$ and applied unchanged

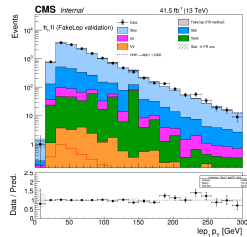
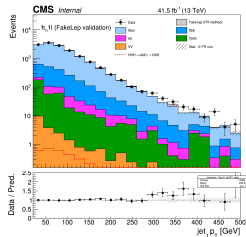
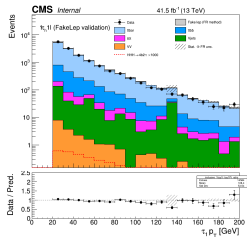
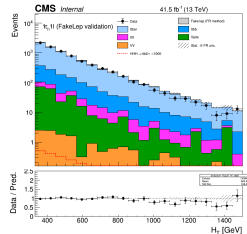
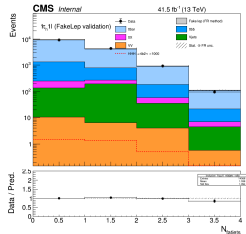
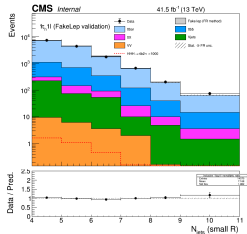


526 events instead of 111 – the higher-statistics test of the same fake factor



$1\tau_h 1l$ with the fake-lepton method only

No data-driven fake- τ template: the τ background comes from MC truth



What the fake-lepton-only configuration shows

- ▶ Turning the data-driven fake- τ template off and keeping only the fake-lepton estimate, the prediction still describes the data across N_{jets} , N_{fatjets} , H_T , p_T^τ , p_T^{jet1} and p_T^ℓ .
- ▶ So in this channel the **fake-lepton contribution is small**, and the **pure-MC description is already good** – the data-driven fake- τ template is a refinement on top of a sound baseline, not a patch covering a mismodelling.
- ▶ This is also why the fake-lepton systematic does not drive the $1\tau_h 1\ell$ uncertainty.

- ▶ MUFFIN reproduces the binned F_F in normalisation (**0.02–0.08%** on the total yield) and improves the **shape**, most visibly in p_T^τ (11.9% $\rightarrow$ 3.8%) and in every SPANet ProbHHH4b2tau training (**1.4–2.7%**).
- ▶ Its systematic is **smaller**: a flat 11.4% per bin against 13.7–16.5% for the map, which degrades into the tails.
- ▶ The cross-channel transfer is validated on $t\bar{t}$ MC: **1.8%** residual for MUFFIN, 3.1% for the binned map.
- ▶ $2\tau_h 0l$: the (tight,tight) region holds 111 events. The same fake factor closes in the 526-event single-tight sideband at **0.998 ± 0.056** , so the method is validated in that channel too.

- ▶ We are applying the full analysis chain to the **other years**, to produce the final complete limit for this version.
- ▶ Each year needs its **own trigger scale factors** and its **own fake factor**, measured separately – that is the bulk of the remaining work.
- ▶ If nothing unexpected comes up, we expect to finish within **one week**.